6. General Methods of Control Available to the Industrial Hygiene Engineer

Introduction

In the previous chapters, the role of the industrial hygiene engineer in recognizing and evaluating the industrial environment has been discussed. This chapter will deal with the general methods of control available to the industrial hygiene engineer when he is faced with the necessity to control a hazard exposure.

The job of the industrial hygiene engineer involves four major functions. These functions are:

A. Recognition of potential hazards in the work environment.

B. Measurement of the work environment to determine the extent of the hazard present and the subsequent evaluation of the measurements obtained to determine if a hazard exists.

C. Identification and recommendation of controls that can be implemented to remove or reduce the hazard exposure of the worker.

D. Anticipation of potential hazards, so these problem hazards can be avoided in the planning stage, if possible.

There are certain principles that relate to the methods that can be used to control hazards. The first principle is that all hazards can be controlled in some manner and to some degree. This is not to say that all hazards are equally easy to control. Some hazard exposures by their very nature present difficult problems to the industrial hygiene engineer. However, through perseverance, exposure to the hazard can be limited. Application of the Golden Rule, "Do unto others as you would have them do unto you," is useful when faced with the temptation to overlook a hazard exposure because of the difficulty required to solve it. As an industrial hygiene engineer, you might put yourself in the worker's shoes: would you be willing to work in the same environment? When you answer this question, you will agree that all hazards can be controlled.

The second general principle relating to the control of hazards is that there are usually many alternate methods of control that are available. As an industrial hygiene engineer, it is your responsibility to identify the alternate methods of control available and to select the most appropriate method. You must resist the temptation to jump to the first solution that is identified.

The third principle is that some methods of control are better than others in a given situation. This principle is almost a corollary to the second principle. In any given situation, there are usually not only many alternate solutions available, but also probably a best solution. The choice is not always easy since there are many factors to consider: the cost of implementation; worker acceptance; enforceability; effect on other operations within the production system; and the continuing operating cost that will be incurred by implementation of the chosen alternative.

The fourth principle of control is that some situations will require more than one control method to obtain optimum results. This is easy to overlook when faced with a problem. Just as there is a tendency to jump to the first solution that is identified, so there is a tendency to use only the single solution chosen as a method of control. Though the chosen solution may improve the situation, it is possible that implementing other alternatives will give greater control of the hazard exposure.

General Methods of Control--Substitution

The first general method of control available to the industrial hygiene engineer is substitution. Substitution can take three forms:

A. Substitution of materials

B. Substitution of process

C. Substitution of equipment

Any one or combination of these forms of substitution may provide a method of control for a given hazard. Referring to the preceding discussion, don't forget that you can substitute using all three forms if necessary to obtain optimum results.

When considering substitution as a method of control, the first question which must be asked is: "Is there a less toxic or flammable material that can do the job?" Given the abundance of materials currently available in the industrial world, substitute material may do the job as well or better than the hazardous material being used. Or it may be necessary to give up some production efficiency to obtain the required control of the hazard exposure to the worker.

Common examples of material substitution include the use of trichloroethylene for carbon tetrachloride and aliphatic chlorinated hydrocarbons for benzene. In each of these cases, the material substituted is less toxic than the original material. In the case of solvents, a further substitution of alkali-and-water detergent solutions may yield equal results with an even greater margin of safety for the worker. Thus, given a particular situation, the material being used might be replaced by another material which involves no hazard exposure.

The second question that might be asked is: "Can the process be changed, thus removing the hazard exposure?" Or worded differently, "Is there a better

way to do the job?" It may be possible to change the overall process or some procedures within the process to eliminate the worker's exposure to hazardous materials or operations. One general principle is usually applicable: a continuous operation is generally less hazardous than an intermittent operation.

For example, when spray-painting a part, consider the possibility of changing to an alternative process such as dipping the part in a paint bath or flow coating it. Either of those processes presents less potential contamination by toxic materials in the air. Another example would be the substitution of automated material handling devices for manual or mechanical methods. This substitution has the additional benefit of eliminating costly manual labor. The substitution of closed system continuous processing for batch processing is an example of the general principle stated above. Where containers of toxic materials must be opened and dumped into a system, consider pumping or conveying the toxic materials from the storage area to the process rather than requiring that they be transported and dumped. Another example is the use of wet methods to reduce dust generation in mining and quarrying operations.

The final type of substitution is substitution of equipment. Is there a better type of equipment to do the job? Can engineering changes be made on the existing equipment to make the equipment less hazardous?

Examples of such substitution include the use of machine guarding on existing mechanical equipment and the substitution of automated equipment for manual methods. Adding the catalytic converter to the automobile to reduce the emission of pollutants is an example of making changes to existing equipment that reduce the hazard potential. The next logical step is to develop an automobile that operates on a nonpolluting fuel: for example, an electric or steam automobile. This approach has led to the use of LPG and electric-powered lift trucks in place of gasoline-powered trucks.

General Methods of Control--Isolation of Source

Another method available for controlling hazard exposures is removing the source of the hazard exposure from the worker's environment. This isolation can be accomplished in a number of ways. First, the source of the hazard exposure can be separated from the work area by placing it in another location where the workers are unlikely to come in contact with it. A second method is to enclose or shield the source with physical barriers. Thus, although the source remains in the work area, it is separated from the workers and the work environment. A third method that was briefly mentioned in relation to substitution of equipment and process is to automate the process so that it operates within a closed system. A fourth method of isolation that controls potential exposure from toxic or flammable materials stored in the production area is the removal and storage of these materials in a separate location.

There are many examples of isolation that have been used in industry. Tank farms that are used for storing toxic or flammable materials in areas apart from the work environment are a type of isolation. The automated processes that are used in chemical processing and petroleum refining are also

examples. Heat barriers and soundproof enclosures have also been used in industry. Another common type of isolation is the removal of the worker to a control room that is separate from the processing area.

General Methods of Control--Ventilation

Ventilation is a useful method for controlling the air quality and the thermal exposures that the worker encounters. Ventilation can be used to remove air pollutants from the breathing zone of the workers. It can also be used to condition the air for worker comfort. In addition, ventilation systems can be designed to supply air to assure the proper operation of any local exhaust system in use.

There is no single type of ventilation system that solves all problems. Ventilation systems come in different sizes and shapes, depending on the requirements of the process and the hazard exposure that is to be controlled. The various categories of ventilation that can be used are:

A. Comfort ventilation. Comfort ventilation moves and conditions the air to assure the comfort of the workers. It is useful in controlling the amount of heat or cold to which the worker is exposed. Also controlled by comfort ventilation are the humidity level and any unpleasant odors that might be present in the work environment.

B. Local exhaust ventilation. Local exhaust ventilation is used to remove contaminants that are generated at a local source. Air is drawn from a source at a rate capable of removing any air contaminants generated at that source before they can be dispersed into the work environment.

C. Local supply ventilation. Local supply ventilation involves supplying air to a specific point or operation where it is required. Local supply ventilation can be used to provide spot cooling for workers in hot areas, or it can be used as one component of a local push-pull ventilation system.

D. Make-up air. Make-up air is air supplied to the work environment to make up for air that is being exhausted through a local system. Without adequate make-up air, local exhaust systems will not operate effectively.

E. Dilution ventilation. Dilution ventilation supplies or exhausts air from a large area in order to control a pollutant in the total area by diluting the pollutant. In general, dilution ventilation is not applicable to highly toxic hazards and proves to be expensive in order to obtain the desired results.

F. Natural ventilation. Natural ventilation systems use the characteristics of air movement created by thermal differences to attain pollution control without mechanical assistance.

The air-conditioning system within an office building, computer room, or factory is an example of comfort ventilation. Such a system provides for the comfort of the worker by adding or removing heat and moisture from the work environment. Exhaust hoods installed over laboratory benches are examples of local exhaust systems. The push-pull ventilation system used over an open tank to capture pollutants before they enter the work environment is an example of a local supply and local exhaust system. The ventilation system that is installed in a traffic tunnel to eliminate carbon monoxide buildup is an example of a dilution ventilation system. Exhausting polluting materials away from a plant through stacks is an example of both natural ventilation and dilution ventilation.

General Methods of Control--Administrative

General administrative controls are those controls available to the organization that do not directly remove the source of hazard exposure from the workplace. These controls are usually effective when used with one of the other control methods previously outlined. Examples of general administrative controls available include worker training, monitoring the work area or the worker, scheduling workers into the area, good housekeeping, and preventive maintenance scheduling to assure proper functioning of the existing controls.

The training of workers has a valuable, though sometimes overemphasized, part in control of hazard exposures. Through training, workers can be taught to identify potential hazards and report these hazards before an incident occurs. Training can be used to provide workers with methods and procedures that are useful in avoiding hazards and to develop error avoidance behaviors in workers.

Another type of administrative control is monitoring the work area or worker. Continuous monitoring equipment can be placed in the work area. This equipment can sound a warning should the potential hazard exceed limits that can become harmful to the workers. Personnel samplers or dosimeters can also be used to monitor the exposure of the worker when he is required to move in and out of the areas where potential hazards exist.

After-the-fact biological monitoring of workers can be of value in determining if a worker has been exposed to a hazard. This biological monitoring can involve pre- and post-employment medical exams to provide data upon which a comparison can be made. Periodic medical examinations should be scheduled for all workers who must work in potentially hazardous areas of the plant. However, the results of such monitoring occur after exposure and may be too late for the worker involved unless the biological monitoring has the sensitivity to identify symptoms before the worker's body is damaged. Therefore, other control techniques should be implemented.

The rotation of worker schedules can control the exposure of any individual worker. Workers can be rotated in and out of hazardous areas during a shift. Workers can be rescheduled to different areas of the plant after a period of time to control possible cumulative effects of the potential hazard. Workers required to perform extremely physical tasks or to work in

hot or cold areas can be given rest periods during which their systems can recover from the exposure. In using this technique caution is required to insure that it does not result in the spreading of serious risk to workers as compared to the institution of permanent corrective action.

Good housekeeping procedures can go a long way to helping control hazards. This includes, but is not limited to, cleanliness of the workplace, waste disposal, healthful drinking water, adequate washing and eating facilities, and control of insects and rodents.

The use of preventive maintenance schedules is a valuable administrative control to eliminate potential hazard exposures. Maintenance on a regularly scheduled basis for potentially hazardous operations is a must. If the worker is to be protected, the system must operate as it was designed to operate. Normal wear can often cause problems to develop that will expose the worker to a hazard that could have been avoided had the equipment been maintained properly. This is also true for any control or monitoring equipment in the workplace. Filters become clogged, fans do not always work as they were designed to work, and monitoring equipment can malfunction.

Other administrative controls that are available include such things as reports and statistics gathered from previous work-related injuries and illnesses and a recognition program that emphasizes regular inspections to identify potential hazard exposures before they become a problem. The existence of adequate emergency aid and emergency procedures can also be used, not as a control of a hazard but as a method to minimize the extent of individual injury and the number of people exposed when an emergency occurs.

General Methods of Control--Personal Protective Equipment

Personal protective equipment should be used only as a last resort and as a temporary measure until more permanent controls can be installed. In somecases there is no immediate alternative but to use personal protective equipment. However, the industrial hygiene engineer should continue to look for other solutions where personal protective equipment is currently being used.

Each type of protective equipment and clothing used should be tested to assure it will do the job for which it was designed. It is also important to be sure the protective equipment and clothing are adequate to provide protection from the hazard. The use of goggles will protect the eyes from damage but will not provide protection for the face. Perhaps a face shield is more appropriate for the potential hazard exposure.

Protective equipment should be designed to provide minimum interference with the job being done. If this is not considered, the worker is likely to discard the protective equipment very quickly and risk being exposed to the hazard. For example, large gloves may interfere with the requirement to perform small psychomotor manipulation of parts. These gloves may be replaced by latex form-fitting gloves that allow the required movements but provide the same protection to the hands.

Personal protective equipment can be categorized as follows:

A. Skin protection--including gloves, suits, and aprons.

B. Eye protection--including safety glasses, goggles, face shields, and hoods.

C. Ear protection--including plugs and ear muffs.

D. Respiratory protection--including air-purifying respirators, air-supplied respirators, and self-contained breathing units.

E. Other protection--including safety shoes, diving suits, and environmental control suits.

Determining the Control Method to Use

The first step necessary to determine the appropriate control method(s) to use for a particular hazard is to be sure that the hazard has been identified correctly. You must know what is being controlled to control it effectively. You should be aware of the entry routes that the hazard may take to the human body. The use of the procedures suggested in Chapter 2 should provide the information that is necessary to establish effective control of the hazard.

After the hazard has been identified and described, it is desirable to identify alternate methods for controlling the hazard. The first method selected as a potential control may not necessarily be the best. Alternate methods should be identified. The problem may be such as to require that a combination of controls be implemented to obtain optimum results. Identification of alternate methods can help to point out the possible application of multiple control methods.

After all the potential alternatives have been identified, it is necessary to determine which alternative(s) provide the best solution to the problem. Which alternative is most effective in controlling the worker's exposure to the hazard? What is the cost of implementing the various alternatives, and which is the most cost effective? What are the ongoing and operational costs in terms of maintenance and other requirements? These questions must be asked to determine which of the alternatives is the most appropriate for implementation.

The best alternative should be selected for implementation. Some of the criteria for selection were mentioned above. Quite often the best alternative involves a trade-off between factors.

The next step is to implement the control(s) that have been selected. This implementation should include a series of tests of the system to assure that, upon operation, the controls will provide the protection for which they were designed.

After the system has been implemented and is operating, periodic follow-up is desirable. Inoperable control mechanisms give a false sense of security

while exposing the worker to a hazard level even higher than that to which he was exposed prior to the implementation of controls. Changes in procedures and processes can make the controls ineffective. Tests on the system are necessary to assure that the controls are doing the job for which they were designed. A proper preventive maintenance schedule should be instituted. Any rules and procedures that govern the workers should be reviewed for compliance, and enforcement must be uniform to assure that the rules and procedures will be followed.

Anticipation of potential hazards is another way of saying that an integral part of the control procedure is to be involved in the planning stages. Proper design of equipment during plant construction is very important in reducing industrial health problems. Provisions must be made for the safe handling of hazardous materials. The health professional should make sure that he is included in the plan-checking procedures. Then he can discover health and safety hazards and correct conditions that would result in hazardous exposures or unsafe practices. Otherwise such conditions may be built into the facility and its equipment.

Many companies have implemented a system requiring that drawings or specifications be approved by the health professional before use. Hazards involved in making products should be minimized insofar as possible. Instructions and warnings developed for employee use should be reviewed for safe manufacturing procedures. The health professional should also make sure that company policies and applicable standards are followed in purchase specifications for new materials and equipment and for modification of existing equipment.

The engineering department, with the help of the health professional, should check with the purchasing department to determine the necessary safety and health measures to be built on or into a machine before it is purchased. The health professional must have as complete a grasp as possible of the occupational disease and accident losses to the company in terms of specific machines, materials and processes. If he is to recommend the expenditure of large sums of money for protection of health to be used throughout the plant, he should have evidence that the investment is justified.

Summary

The industrial hygiene engineer has many methods available for control of hazards in the industrial environment. Using one or more of these methods, it is possible for the industrial hygiene engineer to control to some degree all hazards to which the worker is exposed. The problem becomes one of selecting the control or controls that best provide for the protection of the worker.

The general control methods available to the industrial hygiene engineer include substitution of materials, process, or equipment; isolation of source; ventilation; administrative controls; and personal protective equipment. In many cases, the situation may merit the use of more than one method to provide optimum results in controlling the hazards.

To maximize the potential for success in selecting and implementing the appropriate control method(s), it is desirable that the industrial hygiene engineer use a structured problem-solving approach. Such an approach has been outlined in the chapter.